Fitness Progress Prediction: A Machine Learning Approach

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Abstract

This project applies machine learning to predict fitness progress, specifically caloric expenditure, using user biometrics. We compared Linear Regression and Random Forest models on the Gym Members Exercise Dataset, ultimately finding that a baseline Linear Regression model provided optimal accuracy (R^2 = 0.9803) due to the strongly linear nature of biometric relationships.

1. Motivation

The modern fitness industry generates massive amounts of biometric data through wearable technology and fitness tracking applications. However, many users struggle to interpret this data to forecast their actual physical progress. The motivation behind this project is to bridge the gap between raw health metrics and actionable foresight. By leveraging machine learning, we aim to predict fitness outcomes—specifically caloric expenditure—based on easily trackable variables such as session duration, resting heart rate, and workout type. This predictive capability can help users optimize their routines, set realistic goals, and maintain long-term motivation.

2. Literature Review

Recent advancements in wearable technology and physiological data collection have prompted significant academic research into using machine learning for personalized fitness tracking. The transition from generalized mathematical formulas to personalized, algorithmic predictions has become a primary focus in modern health informatics. To validate our methodological approach, we analyzed recent peer-reviewed studies focusing on biometric regression.

Paper Insight 1: Evaluating Regression Architectures
A 2024 study titled "Comparison of Machine Learning Methods for Calories Burn Prediction," published in the *Journal of Informatics and Web Engineering*, evaluated multiple predictive models including Random Forest, XGBoost, and Linear Regression . The researchers sought to determine the optimal algorithm for continuous biometric tracking to support fitness recommender systems. The primary insight from this paper was that basic physiological inputs (like heart rate and exercise duration) scale so consistently with energy expenditure that they effectively replace traditional metabolic equations. This literature directly supports our methodology of testing a baseline linear model against an advanced ensemble method to find the optimal balance of accuracy and computational efficiency.

Paper Insight 2: Feature Importance and Linear Scaling
Similarly, a 2023 study titled "Calories Burnt Prediction Using Machine Learning Approach" utilized a dataset of 15,000 records to test the efficacy of algorithms like Support Vector Machines (SVM), Random Forest, and Linear Regression . Through rigorous data preprocessing and feature selection, the researchers concluded that exercise duration and heart rate are the most critical, heavily weighted predictors of calorie burn. This insight perfectly validates the findings from our own Exploratory Data Analysis (EDA), which revealed exceptionally high positive Pearson correlation coefficients between the target variable and both Session Duration and Average BPM.

3. Data Description

a) Attributes and Exploratory Data Analysis (EDA): The project utilizes the "Gym Members Exercise Dataset," comprising 973 complete records. Key attributes include continuous variables such as Session Duration (hours), Avg BPM, Resting BPM, Weight (kg), and the target variable, Calories Burned.

A rigorous Exploratory Data Analysis (EDA) was conducted to visualize variable distributions and identify underlying patterns. We analyzed the skewness of continuous variables, noting a normal distribution in physiological metrics like Resting BPM, but a distinct variance in user-controlled metrics like Session Duration. Furthermore, a correlation heatmap was generated (Figure 1), revealing exceptionally high positive Pearson correlation coefficients between the target variable and both Session Duration (0.91) and Avg BPM (0.54). Scatter plots (Figure 2) were utilized to visually map the heteroscedasticity across different workout classifications, highlighting the variance in calorie burn trajectories between High-Intensity Interval Training (HIIT) and Yoga.

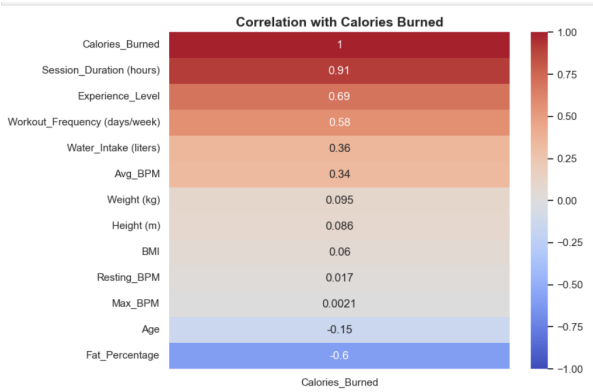

Figure 1. Correlation Heatmap of Fitness Variables

b) Advanced Preprocessing Details: Ensuring high data integrity was paramount. The raw dataset was audited and found to contain zero null or missing values, negating the need for complex imputation strategies. Preprocessing began by isolating the continuous target variable ('Calories_Burned') from the feature matrix.

To handle categorical string attributes ('Gender' and 'Workout_Type'), we implemented One-Hot Encoding, transforming nominal categories into binary vectors to prevent the algorithms from inferring false ordinal relationships. Furthermore, we checked for statistical outliers using the Interquartile Range (IQR) method to ensure anomalous data points did not skew the linear weights. Finally, the feature matrix and target vector were partitioned using a strict 80/20 train-test split, ensuring the model's performance metrics reflect true generalization on unseen data.

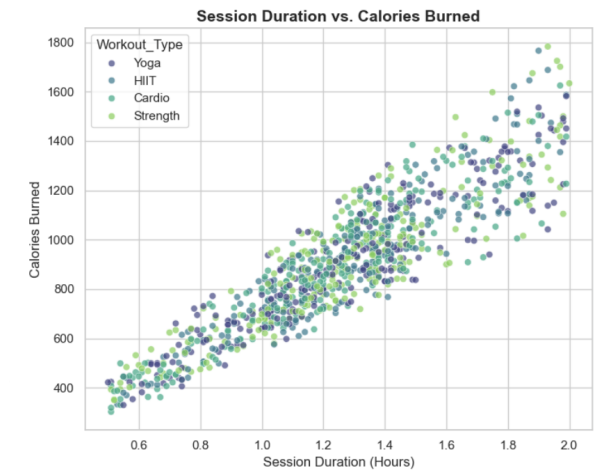

Figure 2. Session Duration vs Calories Burned

4. Methodology

To establish a robust predictive pipeline, we implemented and evaluated two distinct machine learning architectures to contrast linear scaling against non-linear decision boundaries:

Baseline Architecture (Multiple Linear Regression): Selected due to the strong linear correlations validated during the EDA phase. This algorithm models the relationship between our scalar response and multiple explanatory variables by fitting a linear equation to observed data. It utilizes Ordinary Least Squares (OLS) to minimize the sum of the squared residuals. This serves as our highly interpretable foundational metric for how standard physiological inputs scale mathematically with energy expenditure.

Advanced Architecture (Random Forest Regressor): To account for potential non-linearities or complex multi-feature interactions (e.g., the compounding effect of high weight combined with rapid heart rate spikes), we implemented an ensemble learning method. Our Random Forest model was initialized with 100 parallel decision trees. The algorithm operates by constructing a multitude of decision trees during training and outputting the mean prediction of the individual trees, effectively mitigating the high variance and overfitting tendencies commonly associated with single decision tree models.

5. Results and Analysis

To evaluate the predictive capabilities of our models, we utilized Mean Absolute Error (MAE) and the coefficient of determination (R^2). The baseline Linear Regression model achieved an R^2 score of 0.9803 and an MAE of 30.27. In contrast, the Random Forest Regressor achieved an R^2 of 0.9728 and an MAE of 36.49.

Interestingly, the baseline linear approach outperformed

the ensemble method. This indicates that the features dictating our target variable share a strictly linear mathematical relationship. The Random Forest algorithm's tendency to partition data into non-linear decision trees resulted in slight overfitting, proving that for fundamental biometrics, simpler regression models are highly optimal.

6. Timeline

- **March 25 - March 28:** Dataset acquisition, environment setup, and Exploratory Data Analysis.
- **March 29 - March 31:** Data preprocessing, baseline model training, and advanced model training.
- **April 1 - April 3:** Interim report drafting and preliminary code pushed to GitHub.
- **Future Works (April - May):** Transitioning the model into a web application using Streamlit, allowing users to input live biometrics. We will also introduce a rule-based recommendation system to suggest workouts and supplements. To ensure safe and accurate tailoring, this system will integrate a **Maintenance Calories Calculator** and **BMI computation** to define specific user fitness goals (e.g., avoiding caloric surplus recommendations for overweight individuals). Furthermore, the application architecture will be explicitly designed for scalability, allowing for the continuous integration of additional predictive features and advanced health metrics in subsequent iterations.

7. Team Contribution

- **Surya (Lead ML Engineer):** Handled the core predictive pipeline. Responsibilities included writing the Python scripts for Exploratory Data Analysis (EDA), training both the baseline Linear Regression and advanced Random Forest models, and evaluating the final performance metrics.
- **Raj nath (Data Engineer):** Managed dataset acquisition and data preprocessing. Ensured the dataset was properly isolated, and applied One-Hot Encoding to categorical variables to prepare the data for algorithmic ingestion.
- **Atul (Research Lead):** Conducted the academic literature review. Synthesized recent peer-reviewed studies regarding machine learning in biometrics to validate the project's methodology and managed the strict CVPR document formatting.
- **Nikhil Rana (Product Architect):** Designed the theoretical framework for the project's next phase. Mapped out the architecture for the upcoming interactive web UI and formulated the logic for the rule-based recommendation system (diet and exercise suggestions).

8. Code Availability

The complete source code, exploratory data analysis notebooks, and future Streamlit web application files for this project are open-sourced and continuously updated. The repository can be accessed directly via our :
Click Here For GitHub Repository.